Process Characterization Plan

A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) — PCP-008

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

Table of contents

Field	Value
Document ID	PCP-008
Document class	Process Characterization Plan
Title	Process Characterization Plan: Anion Exchange Chromatography (Step 8)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Virology / Viral Safety	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion-exchange chromatography · ANOVA, analysis of variance · CCD, central-composite design · CEX, cation-exchange chromatography · CPP, critical process parameter · CQA, critical quality attribute · CV, column volume · DoE, design of experiments · DS, drug substance · ELISA, enzyme-linked immunosorbent assay · FT, flow-through · GPP, general process parameter · HCP, host-cell protein · icIEF, imaged capillary isoelectric focusing · ICH, International Council for Harmonisation · KPP, key process parameter · LRF/LRV, log-reduction factor / value ·

MSAT, Manufacturing Science and Technology · MVM, minute virus of mice (parvovirus) · NOR, normal operating range · PAR, proven acceptable range · PPQ, process performance qualification · qPCR, quantitative polymerase chain reaction · QbD, Quality by Design · RRF, risk ranking and filtering · RSM, response-surface methodology · SDM, scale-down model · TCID50, 50% tissue-culture infectious dose · UV, ultraviolet absorbance (280 nm) · WC-CPP, well-controlled CPP · XMuLV, xenotropic murine leukaemia virus (enveloped model virus).

1 Purpose and scope

This Process Characterization Plan (**PCP-008**) defines the Stage 1 (Process Design) characterization of the A-Mab anion-exchange (AEX) polishing chromatography step (Step 8), the **flow-through** polishing operation that follows cation-exchange polishing and is the **final chromatographic step** of the purification train. Operated in flow-through (product-transmission) mode, AEX is the step at which the process **establishes the minute-virus-of-mice (MVM, parvovirus) viral-clearance claim** — the CQA it sets — and provides a major further reduction of host-cell protein (HCP), enveloped-virus (XMuLV) clearance, and modular clearance of residual DNA and leached Protein A. The plan specifies the objectives, the attributes in scope, the process parameters and ranges to be characterized, the experimental designs, the analytical methods, the sampling plan, the statistical-analysis approach, and the acceptance and decision criteria by which each parameter will be classified and a multivariate operating region established.

Scope. The commercial-scale anion-exchange polishing step, characterized on a qualified scale-down chromatography column. The upstream cation-exchange step (Step 7), which defines the feed to this step, and the downstream small-virus-retentive filtration step (Step 9), which provides the orthogonal size-based viral clearance, are addressed only where they define the feed to, or complete the clearance credited across, this step; their characterization is out of scope. This plan governs execution only — the results are reported in the paired Process Characterization Report (**PCR-008**).

Basis. The unit-operation model, quality-attribute framework and risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009); the Stage 1 approach and report structure follow the lifecycle approach of the FDA 2011 process-validation guidance (U.S. Food and Drug Administration 2011); the QbD framework follows ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023b, 2012); the viral-clearance framework follows ICH Q5A(R2) (International Council for Harmonisation 2023a).

2 Objectives

The characterization will:

1. establish the quantitative relationships between the anion-exchange process parameters and the flow-through-pool HCP, the enveloped-virus (XMuLV) and parvovirus (MVM) log-reduction, and the step yield;
2. identify the significant main effects, two-factor interactions and curvature, including the anticipated load-pH $\times$ Equil/Wash-1-conductivity interaction that governs HCP clearance and the load-pH/load-conductivity dependence that governs viral clearance;
3. define the multivariate operating region over which the flow-through-pool HCP and the MVM and XMuLV log-reduction credited to this step are simultaneously controlled;
4. establish proven acceptable ranges (PARs) and confirm normal operating ranges (NORs) for each parameter;
5. classify each parameter for its risk of impact to CQAs and to process performance (CPP / WC-CPP / KPP / GPP); and
6. provide the process understanding and ranges that justify the Stage 2 operating conditions and the drug-substance control strategy, including the modular viral-clearance claim.

3 Related documents

This plan is executed under the master documents below and is paired with its report; the full package is cross-referenced for traceability.

Table 3.1: Related documents in the A-Mab process-characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-008	Process Characterization Report (Anion Exchange Chromatography (Step 8))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The governing procedures and validated analytical methods referenced by this plan are listed in Table 3.2 (numbers are placeholders in this synthetic set).

Table 3.2: Referenced standard operating procedures and analytical method validations.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Anion-exchange chromatography is the final polishing step of the A-Mab purification train and is operated in **flow-through** mode: the cation-exchange eluate is conditioned to the target load pH and load conductivity and loaded onto an equilibrated strong anion-exchange resin; at the operating pH the product antibody carries a slight net positive charge and transmits through the column, while the more acidic process-related impurities — host-cell protein, residual DNA and endotoxin — and the model viruses bind to the resin and are retained. The product is recovered in the flow-through together with a post-load wash at the equilibration/wash-1 conductivity, the pool being collected from the start of load to a defined trailing-edge ultraviolet (UV, 280 nm) pool-stop criterion (CMC Biotech Working Group 2009). Like cation exchange it forms no new size- or glycan-related CQA, but it is the step at which the process **credits the MVM (parvovirus) log-reduction** required for the drug-substance viral-safety claim, and it provides orthogonal enveloped-virus (XMuLV) clearance and a major further reduction of HCP with modular clearance of residual DNA and leached Protein A.

Platform knowledge from related humanized IgG1 antibodies established that, in flow-through anion exchange, impurity and virus binding is governed by the electrostatic environment: a **higher load pH** increases the net negative charge of the impurities and virus, strengthening their retention and improving clearance (lower flow-through HCP, higher log-reduction), while a **higher equilibration/wash-1 conductivity** screens those interactions and weakens retention, reducing HCP clearance; the two act together through a load-pH $\times$ conductivity interaction. Viral clearance additionally depends on the **load conductivity**, which must be low enough for the virus to bind. Because the operation is flow-through with high dynamic capacity for the product, the **protein load** can be run high with little effect on clearance or on recovery over the platform range, provided the load material is of representative charge-variant quality; recovery is characteristically high (flow-through). Residence time — set by the operating **flow rate** — governs the contact time available for impurity and virus binding and is held within a range that preserves the minimum contact time credited for the viral-clearance claim. This knowledge frames the attributes in scope and the prioritization of parameters for characterization.

4.2 Quality attributes in scope

Anion exchange **sets one CQA of its own** — the cumulative MVM (parvovirus) log-reduction, to which it is the first and principal contributor — and is a major clearance step for the enveloped-virus (XMuLV) log-reduction and for the process-related impurity CQAs formed upstream and specified at the drug substance: host-cell protein, residual DNA and leached Protein A (Table 4.1). Criticality is scored on the A-Mab continuum using Tool #1 (Risk Score = Impact $\times$ Uncertainty); attributes rated High (H) or Very High (VH) are treated as *Critical* (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b).

Table 4.1: CQAs in scope for the anion-exchange step — the MVM clearance it sets and the XMuLV clearance and impurity CQAs it controls/clears — with acceptance criterion and criticality (Tool #1 = Impact × Uncertainty).

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

The MVM and XMuLV log-reductions are the **cumulative** drug-substance viral-clearance claims (8.6 and 16.7 log respectively), to which the anion-exchange step contributes a modular increment; the flow-through-pool HCP is an in-process attribute whose control at this step determines the margin to the 100 ng/mg HCP drug-substance limit. The MVM and XMuLV log-reduction credited to the step, and the flow-through-pool HCP, are therefore the primary responses of the study; residual DNA and leached Protein A are measured to confirm the modular clearance credited across Protein A, cation exchange and anion exchange.

4.3 Risk-based prioritization of parameters

Consistent with the A-Mab lifecycle of iterative risk assessments, the study scope is set by the Pre-Characterization Process Risk Assessment (**RA-001**). A risk-ranking-and-filtering (RRF) screen scores each parameter for its potential main-effect and interaction impact and assigns the study type accordingly: load pH, equilibration/wash-1 conductivity, load conductivity and protein load — which carry a credible main-effect and interaction risk to the flow-through-pool HCP or to the viral log-reduction — are studied in the multivariate DoE; the operating flow rate, which acts on the viral-clearance claim through residence time rather than through the buffer/load chemistry and is not expected to interact with those factors, is studied univariately across its range and controlled to protect the minimum contact time (CMC Biotech Working Group 2009). The resulting parameter set and study-type assignment are given in Table 6.1.

5 Materials and methods

5.1 Scale-down model and its qualification

Characterization will be performed on a scale-down anion-exchange column operated at the commercial-process equivalents for resin, bed height, residence time, buffer system, load ratio and pool-collection strategy (**SOP-2011**, **SOP-2008**). Prior to characterization the SDM will be qualified against at-scale reference data (**SOP-1001**) by comparing the operational parameters and, critically, the input and output attributes — step yield, flow-through-pool HCP, residual DNA, leached Protein A, and the XMuLV and MVM log-reduction — to confirm the model is representative and suitable to support operating-range, PAR and viral-clearance claims (U.S. Food and Drug Administration 2011; International Council for Harmonisation 2023a). The viral-clearance runs are performed under a dedicated small-scale spiking study using qualified virus stocks. The centre-point replicates of each design provide an in-study estimate of model reproducibility (pure error) used in the lack-of-fit assessment.

5.2 Operation

The cation-exchange eluate is conditioned to the target load pH and load conductivity and loaded onto the equilibrated anion-exchange column; the product transmits in the flow-through while impurities and virus bind; and a post-load wash is applied at the equilibration/wash-1 conductivity. The flow-through/wash pool is collected from the start of load to a defined trailing-edge UV (280 nm) pool-stop criterion, which is a fixed method setting for the study. Buffers, resin lot and column packing are controlled to platform specifications (**SOP-2008**) and are treated as identity/quality-controlled inputs rather than characterized variables (CMC Biotech Working Group 2009). The charge-variant quality of the load material (acidic/deamidated variants) is a controlled input attribute of the feed and is verified prior to loading.

5.3 Analytical methods

The step responses will be measured by the validated methods in Table 5.1; method performance characteristics are per the cited validation reports. HCP is determined by ELISA; the enveloped-virus (XMuLV) and parvovirus (MVM) log-reductions by infectivity titre from the small-scale spiking study; residual DNA by qPCR; leached Protein A by ELISA; and the load-material and pool charge-variant (acidic/deamidated) profile by imaged capillary isoelectric focusing (icIEF).

Table 5.1: Validated analytical methods for the characterization.

Reference	Title	Type
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

5.4 Sampling plan

Each DoE run will be sampled at the flow-through pool for HCP, residual DNA and leached Protein A, and the load and pool product mass will be recorded for the step-yield determination; the paired viral-spiking runs provide the XMuLV and MVM log-reduction. The load-material acidic-variant profile is recorded for each execution. In-process readings (pH, conductivity, UV) will confirm the chromatographic profile and the pool collection window. Centre-point runs are replicated within each design (three in screening, four in the response-surface design) to quantify reproducibility and pure error.

5.5 Statistical methods

Designs will be constructed and analyzed in coded factors, with each parameter's characterization range mapped so that the set-point is 0 and the range edges are ±1 (**SOP-1002**). Screening data will be analyzed by ordinary-least-squares regression of each response on the main effects and all two-factor interactions; factor effects are reported as twice the coded coefficient. Response-surface data will be analyzed with the full quadratic model (main effects, two-factor interactions and pure-quadratic terms). Significance will be assessed at $= 0.05$. Model adequacy will be judged by R^2, adjusted R^2, predicted R^2 (from the PRESS statistic), the overall model F-test, and an ANOVA lack-of-fit test against the centre-point pure error; a response with no significant parameter effect beyond a single dominant factor will be reported as robust rather than modelled predictively. The operating region will be defined as the multivariate region of the well-controlled CPPs over which the flow-through-pool HCP and the credited MVM and XMuLV log-reduction are controlled, and commercial-scale capability of the drug-substance CQAs governed by the step will be estimated by Monte-Carlo simulation with each parameter varied within its NOR.

6 Study design

6.1 Factors, ranges and study type

Five parameters are in scope (Table 6.1). The characterization range of each factor (“Range studied”) defines the edges of the explored knowledge space and the basis of the PAR; the NOR is the tighter routine-control range. Classification is an output of the study and is reported in PCR-008. The DoE factor coding for the four multivariate factors is given in Table 6.2.

Table 6.1: Anion-exchange parameters, set-points, characterization ranges and planned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Load pH	pH	7.5	7.2–7.8	7.35–7.65	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	1.6–3.6	2.1–3.1	multivariate
Load conductivity	mS/cm	5.5	3–8	4–7	multivariate
Protein load	g/L resin	200	50–300	120–280	multivariate
Operating flow rate	cm/hr	300	150–450	200–400	univariate

Table 6.2: Design-of-experiments factor legend (coded A–D).

Code	Factor	Unit	Studied in
A	Load pH	pH	screening + RSM
B	Equil/Wash-1 conductivity	mS/cm	screening + RSM
C	Load conductivity	mS/cm	screening + RSM
D	Protein load	g/L resin	screening + RSM

6.2 Screening design

Factor screening will use a two-level full factorial in the four multivariate factors (A–D) with three centre-point replicates, 19 runs in total. The full factorial estimates all four main effects and all six two-factor interactions without confounding. Four responses will be recorded for each run (flow-through-pool HCP, XMuLV log-reduction, MVM log-reduction and step yield). The planned coded design is given in Appendix A.

6.3 Response-surface design

The four multivariate factors will be carried into a face-centred central-composite design with four centre-point replicates, 28 runs in total, enabling estimation of curvature in addition to main effects and interactions. The planned coded design is given in Appendix B.

6.4 Univariate assessment

The operating flow rate will be evaluated one factor at a time across its characterization range for its effect on the flow-through-pool HCP, the XMuLV and MVM log-reduction and the step yield; it will not be carried into the multivariate design because it acts on viral clearance through residence time rather than through the buffer/load chemistry and is not expected to interact with the chromatographic factors. It is controlled as a well-controlled CPP to preserve the minimum residence time credited for the viral-clearance claim.

6.5 Pool-collection criterion

The flow-through/wash pool is collected to a fixed trailing-edge UV (280 nm) pool-stop criterion applied identically to every run so that the factor effects are estimated free of run-to-run pooling variation. The adequacy of the pool-stop set-point — the balance between flow-through-pool HCP and step yield — will be evaluated as part of the study and confirmed in PCR-008.

7 Acceptance and decision criteria

The study is acceptable when:

1. the SDM qualification confirms representativeness of commercial scale (**SOP-1001**) and the viral-spiking model is qualified;
2. the fitted response-surface models for the flow-through-pool HCP and the MVM and XMuLV log-reduction are adequate — target R^2 0.90, adjusted R^2 0.75, no significant lack of fit ($p > 0.05$ against the centre-point pure error), and residual diagnostics without systematic structure (a response governed by a single dominant factor is reported by that dependence and is otherwise robust); and
3. a multivariate operating region exists over which the flow-through-pool HCP and the MVM and XMuLV log-reduction credited to the step are controlled at the set-point and across each parameter's NOR.

Each parameter will then be classified per the A-Mab continuum (CMC Biotech Working Group 2009):

- a parameter whose variability significantly impacts a CQA is a **CPP**; where the impact is reliably controlled within the operating region (low risk of leaving it) the parameter is a **well-controlled CPP (WC-CPP)**;
- a parameter that does not significantly impact a CQA but affects process performance or consistency is a **KPP**; otherwise it is a **GPP**;
- consistent with the post-characterization FMEA convention, a parameter that affects a CQA with Severity 8, or that carries an initial (pre-mitigation) risk priority number > 72, is designated a CPP before mitigation by the control strategy.

The operating region, classifications and PARs established here will justify the Stage 2 operating ranges and feed the drug-substance control strategy, including the modular viral-clearance claim.

8 Data management and integrity

All raw data, DoE designs, analytical results, viral-spiking records and statistical analyses will be recorded and retained under the site data-integrity procedures (ALCOA+). Analyses will be performed with version-controlled scripts and reviewed by a second analyst; designs, models and the derived operating region will be archived with the report. Any deviation from this plan will be documented and assessed for impact in PCR-008.

9 Roles and responsibilities

Table 9.1: Roles and responsibilities.

Function	Responsibility
Process Development / MSAT	Author the plan; design, execute and analyze the DoE; author PCR-008
Analytical Development	Perform released and in-process testing under the referenced AMV methods
Virology / Viral Safety	Execute and interpret the small-scale viral-spiking clearance study
Manufacturing Science	Qualify the scale-down model; confirm commercial-scale representativeness
Statistics	Review the design, the statistical analysis and the operating-region definition
Quality Assurance	Approve the plan and the report; confirm parameter classifications

10 Deliverables and schedule

The deliverable of this plan is the Process Characterization Report (**PCR-008**), reporting the executed designs, the significant effects, the operating region, the parameter classification, the modular viral-clearance credit and the impurity-CQA outcomes. PCR-008 rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment. Execution follows SDM qualification and precedes the Stage 2 PPQ campaign.

11 Risks and assumptions

The plan assumes the SDM and the viral-spiking model are representative of commercial scale (confirmed by qualification) and that the platform analytical methods are validated and in control. It further assumes that the load material presented to the step is of representative charge-variant (acidic/deamidated) quality; an atypical load — for example one carrying an elevated deamidated-variant burden — could alter the anion-exchange binding balance and is a recognized risk to the characterization. The principal risks are therefore that the absolute clearance shifts at commercial scale or over the resin life-cycle, that the incoming impurity or charge-variant load is higher than characterized, or that the pool-collection UV set-point is not optimal; these are mitigated by defining the operating region on the qualified SDM, verifying the load-material charge-variant quality before loading, confirming the region and the clearance at scale in Stage 2, and carrying the WC-CPP ranges and the pool-collection criterion forward with in-process monitoring and resin life-cycle control.

12 Approvals

Approval of this plan authorizes execution of the characterization described herein. Signatures are recorded in Table 1.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.

International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.

International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.

Appendix A — Planned screening design

Coded two-level full-factorial design (factors A–D per Table 6.2; coded levels $-1/0/+1$). Responses will be recorded at execution and reported in PCR-008.

Table 12.1: Appendix A — planned screening design (coded factor settings).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Appendix B — Planned response-surface design

Coded face-centred central-composite design (factors A–D; coded levels $-1/0/+1$).

Table 12.2: Appendix B — planned response-surface design (coded factor settings).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0